
Structural Reconfiguration of AI Judgment Bias under Information Loss in 6G Communication Environments

GPT-5.1, Gemini 3 Pro and DeepSeek V3.2
(Names with versions only, sorted by priority)*

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Abstract

Sixth-generation (6G) communication technology, unlike previous 5G systems, does not primarily aim to transmit all available information in a maximally compressed form. Instead, 6G is oriented toward an information-processing structure in which only core information necessary for judgment is selectively organized and transmitted at ultra-high speed. In such environments, artificial intelligence systems are more likely to perform judgments based on summarized and refined information rather than directly processing large volumes of original text. As a result, information loss becomes an inevitable component of the judgment process. The critical issue is that this information loss does not merely reduce the amount of information available, but has the potential to alter the composition of judgment inputs themselves, thereby structurally amplifying judgment bias toward particular perspectives.

1 Introduction

1.1 Reframing 6G Communication: From Transmission to Judgment-Oriented Information Processing

Sixth-generation (6G) communication technology does not aim simply to transmit larger volumes of information at higher speeds when compared to previous 5G systems. Rather, 6G is oriented toward an information-processing structure that selectively organizes and transmits only the core information necessary for judgment within ultra-high-speed and ultra-low-latency environments. In such contexts, artificial intelligence functions not as an entity that directly processes extensive original texts, but as a central decision-making agent that forms judgments based on summarized and refined information. While this shift enhances the efficiency of AI judgment, it simultaneously introduces a new structural problem in the form of information loss.

1.2 Information Loss and the Emergence of Structural Judgment Bias

Information loss arising from summarization and compression inevitably weakens or removes the detailed context, subsidiary arguments, and the strength of opposing perspectives contained in original texts. In judgment tasks involving controversial issues, which information is selected as “core” and which is excluded plays a decisive role in shaping outcomes. Consequently, summarized

information environments structurally increase the likelihood that judgment inputs become skewed toward particular perspectives when compared to original-text environments. This dynamic may function as a new mechanism through which AI judgment bias emerges.

1.3 Limitations of Mean-Based Fairness Evaluation in AI Judgment

Existing studies on AI judgment bias have primarily focused on data imbalance or algorithmic design, emphasizing whether average judgment outcomes shift in a particular direction. Although such approaches are effective in capturing directional bias, they are limited in their ability to explain how judgment responses are structurally formed—specifically, how judgment stability and diversity change. Even when judgment outcomes share identical mean values, cases in which judgments are distributed across diverse perspectives and cases in which they converge strongly toward a single value represent fundamentally different judgment structures.

Meanwhile, real-world information environments are rarely balanced. In online information consumption contexts, echo chamber phenomena—where specific perspectives are repeatedly exposed—are commonly observed. Prior research has shown that such repeated exposure can reinforce judgments and intensify polarization (Bail et al.; Cinelli et al.). AI-based information summarization in 6G environments may reproduce these conditions in an even more compressed form. When information input imbalance is combined with information loss, judgment bias may be structurally amplified beyond simple shifts in average outcomes.

1.4 Research Objectives and Contributions

Based on this problem awareness, the present study assumes a 6G-based information-processing environment and seeks to experimentally analyze the structural effects of information input ratios and information representation formats on AI judgment bias. Specifically, the study manipulates the ratio of pro and con claims under balanced and dominant conditions, while comparing original-text inputs and summarized-text inputs under identical information ratios to disentangle the effects of information loss and information imbalance.

Furthermore, by evaluating judgment fairness using a dual-metric framework based on both mean and variance rather than a single mean value, this study proposes an analytical approach that simultaneously considers judgment directionality and the stability and diversity of judgment structures. Through this analysis, the study aims to conceptualize AI judgment bias in 6G environments not as a simple outcome-level deviation, but as a problem of reconfiguration in information input structures and judgment spaces. These findings provide important theoretical and experimental implications for the design and evaluation of AI judgment systems in next-generation communication environments.

2 Experimental Design and Methodology

2.1 Research Overview and Experimental Objectives

The objective of this study is to analyze how information loss influences the structural formation of judgment bias in artificial intelligence systems when making decisions on controversial issues. This study is grounded in the premise that evaluating judgment fairness solely through mean values is insufficient for capturing the full characteristics of AI judgment behavior. Accordingly, the study adopts an analytical framework that simultaneously considers judgment directionality and judgmental structural stability and diversity through the combined use of mean and variance.

To achieve this objective, three key experimental factors were systematically manipulated: (1) the balance ratio of opposing claims in the input data, (2) the mode of information representation (original text versus summarized text), and (3) the nature of the judgment domain. Under each experimental condition, judgment responses generated by AI models were collected and analyzed using mean and variance as the primary indicators. This design enables the detection of structural bias and convergence phenomena within the judgment space, even in cases where surface-level

mean values remain close to neutrality.

2.2 Selection of Judgment Topics and Domain Configuration

The judgment topics employed in this experiment were organized into three distinct domains with differing characteristics to examine whether AI judgment bias manifests differently depending on domain attributes. Topic selection followed three criteria: the presence of a clearly dichotomous pro-con structure, social and technological relevance, and sufficient differentiation to allow comparative analysis.

Based on these criteria, three domains were selected. The first domain concerns maximizing artificial intelligence efficiency versus prioritizing transparency and stability. This domain represents a technical and self-referential debate in which AI systems are required to evaluate principles directly related to their own design and operation. The second domain addresses the expansion of nuclear power plants, combining technical evaluation with policy-oriented and societal considerations. The third domain focuses on the necessity of human intervention in ecosystems, representing socio-environmental controversies in which ethical and value-based judgments play a dominant role.

This configuration provides a theoretical foundation for interpreting domain-specific differences in judgment stability and summarization effects observed in subsequent analyses.

2.3 Information Input Ratios and Representation Formats

To analyze the effects of information loss on AI judgment bias, this study identified information input ratio and information representation format as core experimental variables. The information input ratio refers to the relative exposure frequency of pro and con claims within the judgment input environment. An imbalanced ratio of 8:2 was designed to simulate repeated exposure to a dominant perspective rather than extreme single-sided information exposure. This setting reflects realistic information environments in which claim imbalance accumulates over time, reinforcing bias. Such conditions are analogous to echo chamber phenomena observed in online information consumption contexts, where repeated exposure to homogeneous viewpoints strengthens existing judgments and amplifies polarization.

By contrast, a balanced 5:5 ratio was established as a baseline condition for comparative analysis. Under this condition, AI systems are theoretically more likely to produce neutral judgments, allowing structural changes in mean and variance to be examined independently of input imbalance. Importantly, reductions in variance or convergence observed even under balanced conditions indicate the presence of internal structural bias beyond simple shifts in information ratios. Regarding information representation, original-text inputs and summarized-text inputs were compared under identical ratio conditions. This distinction reflects anticipated information-processing characteristics of 6G environments, where ultra-high-speed and ultra-low-latency communication necessitates the use of AI-based summarization and refinement. Summarized texts preserve central claims while reducing contextual and argumentative detail, potentially reshaping judgment input structures and influencing both judgment diversity and stability.

2.4 Selection of Artificial Intelligence Models

Two commercial large language model-based AI systems with distinct functional roles were selected to examine judgment structure changes under information loss conditions. This role differentiation avoids interpretations driven by the characteristics of a single model and enables a more generalized analysis. ChatGPT was designated as the AI responsible for summary-based information processing. Its capacity to reconstruct long texts around core claims aligns closely with the information summarization and refinement functions expected of AI systems operating in 6G environments. Gemini Pro was employed as the AI responsible for judgment formation and

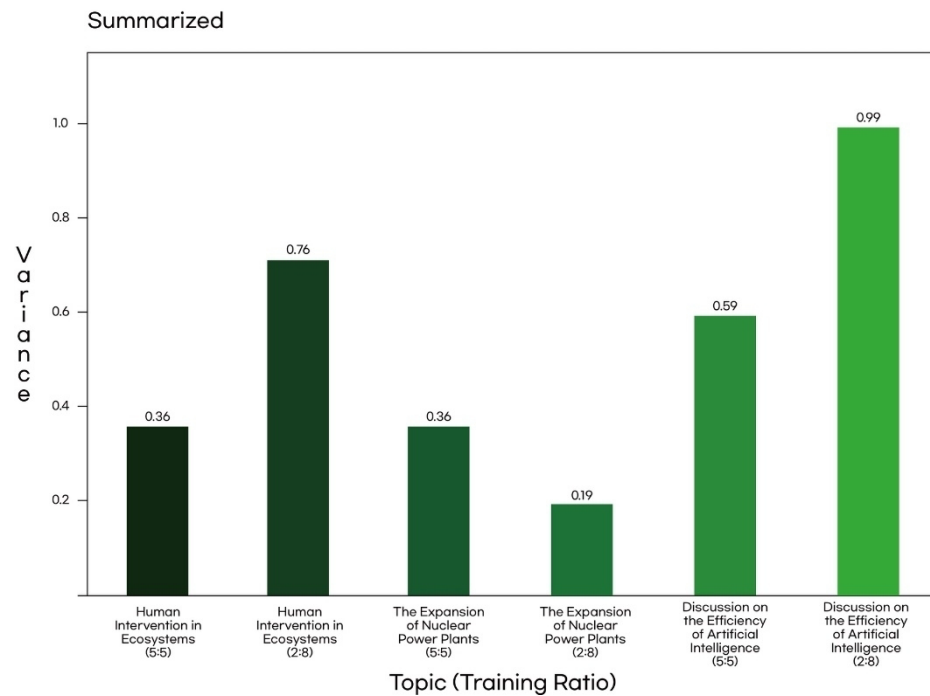
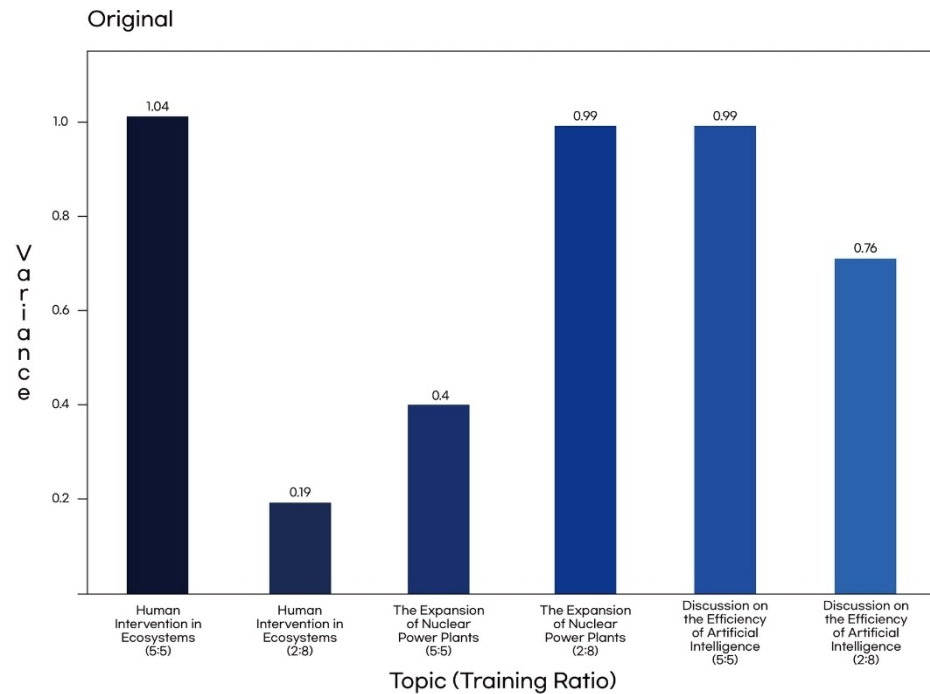
bias analysis. Its inference-oriented characteristics allow it to respond sensitively to variations in input distribution and contextual emphasis, making it suitable for observing changes in judgment directionality and variance. Open-source models were not included, as the study aims to analyze judgment structures in AI systems currently deployed in realistic information environments. Commercial models are more likely to be adopted in 6G contexts due to performance, stability, and real-time requirements, allowing the findings to be interpreted as implications for real-world AI judgment behavior.

2.5 Evaluation Method: Dual-Metric Design Based on a Likert Scale

A five-point Likert scale was employed to quantitatively analyze AI judgment bias. This scale captures graded judgment intensity, including neutral responses and subtle variations, enabling analysis beyond binary outcomes. Mean values were used to represent judgment directionality, while variance was used to assess judgmental stability and diversity. This dual-metric design allows identification of cases in which judgments converge strongly or fluctuate under identical conditions, even when mean values remain close to neutrality. Accordingly, judgment fairness is conceptualized not as a simple central tendency, but as a structural property of the judgment space itself.

3 Experimental Results

Topic	Text Type	Training Ratio	Mean	Variance
Human Intervention in Ecosystems	original	5:5	2.6	1.04
Human Intervention in Ecosystems	original	2:8	3.1	0.19
Human Intervention in Ecosystems	summarized	5:5	3.2	0.36
Human Intervention in Ecosystems	summarized	2:8	3.2	0.76
The Expansion of Nuclear Power Plants	original	5:5	3.0	0.4
The Expansion of Nuclear Power Plants	original	2:8	2.9	0.99
The Expansion of Nuclear Power Plants	summarized	5:5	3.2	0.36
The Expansion of Nuclear Power Plants	summarized	2:8	3.1	0.19
Discussion on the Efficiency of Artificial Intelligence	original	5:5	3.1	0.99
Discussion on the Efficiency of Artificial Intelligence	original	2:8	2.8	0.76
Discussion on the Efficiency of Artificial Intelligence	summarized	5:5	3.1	0.59
Discussion on the Efficiency of Artificial Intelligence	summarized	2:8	2.9	0.99



3.1 Judgment Outcomes under Balanced Input Conditions (5:5)

Under balanced input conditions, mean judgment values across all three domains remained close to the neutral midpoint of the Likert scale. This pattern was observed in both original-text and summarized-text conditions, suggesting that balanced exposure limits overt directional bias at the aggregate level. However, variance patterns differed notably across information representation formats. In summarized-text conditions, variance tended to decrease relative to original-text conditions, indicating increased convergence in judgment responses. This reduction occurred even when mean values remained neutral, suggesting that summarized inputs influence judgment structure independently of directional outcomes.

3.2 Judgment Outcomes under Biased Input Conditions (8:2)

Under biased input conditions, directional shifts in mean values were observed across domains, reflecting sensitivity to dominant claim exposure. In these conditions, summarized-text inputs often intensified convergence effects, either suppressing variance or reorganizing judgment distributions into narrower ranges. Importantly, variance behavior was not uniform across domains. In some cases, summarized inputs amplified instability, producing higher variance despite similar mean values. These findings indicate that information loss interacts with input imbalance in domain-dependent ways, reshaping judgment structures rather than simply shifting outcomes.

3.3 Effects of Information Representation on Judgment Structure

Comparisons between original-text and summarized-text inputs under identical ratio conditions revealed that information representation significantly influences judgment structure. Summarized inputs frequently reduced judgment diversity by limiting the range of interpretive pathways available to the AI system. These effects were observable even in balanced conditions, demonstrating that information loss alone can induce structural convergence. Thus, summarization does not merely compress information volume, but actively reorganizes the configuration of judgment inputs.

3.4 Domain-Specific Differences in Structural Bias

Domain-specific patterns emerged across the three judgment topics. The AI efficiency domain exhibited strong convergence tendencies, reflecting the self-referential nature of the judgment task. The nuclear power domain showed moderate sensitivity to input imbalance, combining technical and societal considerations. The ecosystem intervention domain displayed pronounced structural effects, including cases of strong convergence under both balanced and biased conditions. These differences indicate that judgment bias formation is contingent on domain characteristics, particularly the degree to which ethical, technical, or self-referential considerations are involved.

4 Conclusion

4.1 Summary of Key Findings

This study reconceptualized artificial intelligence judgment bias in 6G environments not as a simple skew in outcomes, but as a structural transformation of the judgment space. The experimental results showed that summarized information inputs, even when maintaining mean judgment values close to neutrality, reduce judgment variance or reorganize it into specific forms, thereby limiting the range of interpretations that artificial intelligence can consider. This indicates that AI judgment bias does not necessarily appear only as extreme directional shifts, but may be concealed in structural forms in which judgment diversity and flexibility are suppressed.

In particular, the results of this study clearly show that apparent neutrality does not necessarily imply fairness. Even when judgments remain neutral on average, excessive reductions in judgment variance may indicate an environment in which diverse perspectives and alternative interpretations are excluded. That is, neutrality observed in highly compressed information environments based on 6G may not be an indicator of fairness, but rather a signal that the judgment space has become excessively standardized.

4.2 Redefining AI Judgment Fairness in the 6G Era

Such structural bias is directly connected to the core characteristics of 6G communication environments. Rather than aiming for the maximum transmission of information, 6G is oriented toward an environment that selectively extracts, summarizes, and structures only the core information necessary for judgment and action, and delivers it with ultra-low latency. In this process, information summarization and refinement inevitably occur in advance, and the results of this study experimentally show that AI judgment bias can be formed not merely as a problem of judgment algorithms, but already at the stage where input information is configured.

In other words, fairness issues in 6G environments are embedded not after judgment, but prior to judgment, within the information-processing structure itself. This finding reframes AI judgment fairness as a problem of information configuration and judgment space design, rather than solely as an issue of algorithmic output.

4.3 Implications for AI Co-Scientist System Design

To analyze these issues, this study utilized artificial intelligence not simply as an object of analysis, but as a role-differentiated research collaborator. Specifically, one AI model was responsible for information summarization and structuring, while another AI model was designed to form judgments based on those inputs, thereby experimentally implementing an inter-AI collaborative structure anticipated in real 6G environments. Through this approach, the study demonstrates how a collaborative research structure—where human researchers lead experimental design and interpretation while artificial intelligence systems divide roles in information processing and judgment execution—contributes to the analysis of judgment bias.

The results derived from this collaborative structure provide important implications for the design of AI Co-Scientist systems. When the AI responsible for summarization and the AI responsible for judgment are integrated into a single structure, convergence effects resulting from information loss may be invisibly intensified. Therefore, future AI collaborative systems require mechanisms that monitor not only the mean values of judgment outcomes but also judgment variance and structural stability. Design strategies that manage summarization sensitivity in a domain-specific manner are expected to be necessary.

This implies that a Co-Scientist should not merely function as a tool for efficiently producing conclusions, but should instead serve as a research partner that actively manages and audits the structure of judgment itself.

4.4 Limitations and Future Research Directions

Finally, this study conducted its analysis using limited domains, a single summarization method, and commercial large language models. Future research will need to analyze more precisely how AI judgment structures are formed and adjusted in real-world, 6G-scale information environments through expanded collaborative designs that incorporate diverse summarization strategies and inter-AI feedback structures.

Nevertheless, this study holds important significance in that it redefines AI judgment fairness in the 6G era not as a matter of outcomes, but as a structural issue, through collaboration between artificial intelligence and humans.

References

- [1] Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. B. F., Lee, J., Mann, M., Merhout, F., & Volfovsky, A. (2018). Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37), 9216–9221. <https://doi.org/10.1073/pnas.1804840115>
- [2] Cinelli, M., De Francisci Morales, G., Galeazzi, A., Quattrociocchi, W., & Starnini, M. (2021). The echo chamber effect on social media. *Proceedings of the National Academy of Sciences*, 118(9), e2023301118. <https://doi.org/10.1073/pnas.2023301118>
- [3] OpenAI. (n.d.). ChatGPT capabilities overview. OpenAI Help Center. <https://help.openai.com/>
- [4] Google DeepMind. (n.d.). Gemini 3 Pro. Google DeepMind Developer Documentation. <https://deepmind.google/>

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Justification: The study was conducted entirely through AI-to-AI interaction models and did not involve any human participants or crowdsourced labor.

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